

BATANG, China

WMO: 562470

Lat: 30.0029N

Lon: 99.1077E

Elev: 8494

StdP: 10.71

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.5	25.9	-8.9	4.7	43.0	-5.6	5.6	43.9	13.0	58.5	11.3	57.5	1.0	160	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	22.5	89.1	58.8	86.2	58.4	83.5	58.0	63.5	77.2	62.5	76.2	61.6	74.8	5.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.1	107.2	66.2	59.1	103.4	65.3	58.2	100.0	65.0	34.0	77.9	33.1	76.4	32.3	75.0	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.3	9.3	DB	18.5	94.4	2.0	2.5	17.1	96.2	15.9	97.7	14.8	99.1	13.3	100.9
			WB	13.0	65.6	1.6	1.2	11.9	66.5	10.9	67.2	10.0	67.9	8.9	68.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.5	42.6	47.2	52.0	57.9	65.4	70.3	69.9	69.0	65.4	58.2	49.1	42.5
	DBStd	11.14	4.55	4.83	4.39	4.50	5.45	5.34	4.65	4.07	4.53	4.83	4.13	4.14
	HDD50	666	233	102	28	3	0	0	0	0	0	2	63	235
	HDD65	3348	695	499	403	216	63	13	11	12	47	215	476	698
	CDD50	3408	4	24	90	240	478	608	616	588	463	258	37	2
	CDD65	612	0	0	0	3	76	171	162	135	60	5	0	0
	CDH74	5307	0	1	11	122	934	1557	1145	907	505	124	1	0
	CDH80	1539	0	0	0	12	247	571	353	237	112	7	0	0
(14) Wind	WSAvg	2.8	2.8	3.7	4.0	3.7	3.5	3.0	2.3	2.3	2.0	2.4	2.2	2.2
(15) Precipitation	PrecAvg	18.6	0.0	0.0	0.2	0.6	1.3	3.5	4.9	3.9	3.4	0.8	0.1	0.0
	PrecMax	27.0	0.0	0.2	0.5	1.9	3.9	6.2	13.3	8.3	6.7	3.0	0.9	0.4
	PrecMin	9.1	0.0	0.0	0.0	0.1	0.1	0.9	1.8	0.7	1.2	0.0	0.0	0.0
	PrecStd	4.3	0.0	0.0	0.2	0.4	0.8	1.5	2.3	2.0	1.3	0.6	0.2	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.9	71.5	75.5	81.6	89.1	93.6	92.2	89.5	87.8	80.7	72.0	65.3
		MCWB	41.2	43.0	46.5	51.4	56.3	59.4	60.9	59.7	59.0	54.3	46.3	40.8
	2%	DB	62.0	67.1	71.2	77.3	85.7	89.7	87.6	85.5	83.0	77.3	68.5	61.4
		MCWB	38.7	41.4	44.5	49.4	55.0	58.7	60.6	59.9	58.4	53.2	44.4	39.3
	5%	DB	58.6	63.7	68.3	74.2	82.7	86.4	83.8	82.4	79.8	74.5	65.6	58.4
		MCWB	37.1	40.0	43.3	48.1	54.0	58.1	60.5	59.9	58.1	52.0	43.6	37.9
	10%	DB	55.3	60.5	65.3	71.3	79.5	82.9	80.4	79.2	76.3	71.2	62.5	55.5
		MCWB	35.6	38.7	42.2	47.1	53.1	57.7	60.2	59.8	57.7	50.4	42.2	36.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.9	44.1	48.2	53.5	59.1	63.8	65.4	64.6	62.8	57.3	48.6	42.0
		MCDB	64.7	67.6	69.8	76.8	81.1	79.2	77.8	78.0	76.7	74.7	66.5	62.6
	2%	WB	39.4	42.1	45.9	51.1	56.8	62.2	63.7	63.1	61.2	55.6	46.4	39.9
		MCDB	60.6	64.3	66.9	72.0	78.2	78.1	77.2	76.3	74.7	72.7	63.9	59.6
	5%	WB	37.5	40.5	44.3	49.6	55.6	61.1	62.7	62.1	60.1	54.2	44.6	38.4
		MCDB	57.4	61.6	64.4	69.3	77.0	76.5	76.4	74.8	73.1	69.4	62.4	57.1
	10%	WB	35.9	39.1	43.1	48.3	54.4	60.1	61.7	61.2	59.0	52.6	43.0	36.8
		MCDB	54.6	59.2	62.4	67.2	75.0	74.7	74.6	73.3	71.6	66.2	60.1	54.4
(35) Mean Daily Temperature Range	5% DB	MDBR	28.7	27.7	26.0	25.4	25.2	22.5	20.5	20.5	22.0	26.9	29.4	29.2
		MCDBR	32.3	32.5	31.1	30.8	30.1	28.2	26.6	26.7	28.3	30.8	33.0	32.0
		MCWBR	16.5	15.7	13.4	11.4	9.6	7.6	6.7	7.4	8.2	10.8	14.9	16.2
	5% WB	MCDBR	31.3	30.5	27.1	27.4	25.9	22.0	20.8	20.4	21.3	26.0	30.0	31.0
		MCWBR	16.3	14.9	11.4	10.3	8.6	6.5	6.4	6.6	7.3	9.3	14.0	15.9
(40) Clear-Sky Solar Irradiance	taub		0.258	0.298	0.374	0.389	0.351	0.337	0.347	0.351	0.320	0.279	0.254	0.242
	taud		2.504	2.395	2.152	2.171	2.390	2.534	2.521	2.492	2.556	2.620	2.598	2.583
	Ebn at Noon		315	308	288	287	297	299	296	295	303	312	315	318
	Edh at Noon		28	34	47	48	38	33	34	34	31	27	25	24
(44) All-Sky Solar Radiation	RadAvg		1326	1531	1755	1892	1967	1900	1763	1718	1601	1480	1362	1257
	RadStd		73	95	142	141	124	163	159	162	110	87	81	54

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.12	+1.15	-2.92	+2.14	N/A	N/A	-108	-238	+340
(47) Regional (4 neighbors)		+0.92	+0.90	-1.75	+1.39	N/A	N/A	-146	-299	+180

Nomenclature: See separate page